EMT 1101 *Continuous Assessment Examination Test*

Friday, December 19, 2014 **Time:** $1\frac{1}{2}$ **hours**

1. Check that the paper has **9 questions**.
2. Attempt all questions
3. Write answers in the space provided
4. Show all your working clearly
5. Use proper notation
6. Simplify your answers as far as possible
7. **No Calculators**

Name _____

Reg Number , Programme: _____

Question:	1	2	3	4	5	6	7	8	9	Total
Points:	5	5	4	9	7	9	4	6	10	59
Score:										

1. Simplify the following:
   (a) $4\sqrt{24} - 8\sqrt{54} + 2\sqrt{6}$ (2)

   Solution: $4\sqrt{24} - 8\sqrt{54} + 2\sqrt{6} = 8\sqrt{24} - 24\sqrt{6} + 2\sqrt{6} = -12\sqrt{6}$

   (b) $\sqrt[5]{\frac{\sqrt[5]{x}\sqrt[4]{y}}{\sqrt[4]{z^2}}}$ leaving your answer without the outer radical sign. (3)

   Solution: $\sqrt[5]{\frac{\sqrt[5]{x}\sqrt[4]{y}}{\sqrt[4]{z^2}}} = \left(\frac{\sqrt[5]{x}\sqrt[4]{y}}{\sqrt[4]{z^2}}\right)^{\frac{1}{5}} = \frac{\left(\sqrt[5]{x}\right)^{\frac{1}{5}}\left(\sqrt[4]{y}\right)^{\frac{1}{5}}}{\left(\sqrt[4]{z^2}\right)^{\frac{1}{5}}} = \frac{\sqrt[25]{x}\sqrt[20]{y}}{\sqrt[20]{z^2}}$

2. Factorise
   (a) $x^6 - y^6$ as a difference of cubes (2)

   Solution: $x^6 - y^6 = (x^2)^3 - (y^2)^3 = (x^2 - y^2)((x^2)^2 + y^2x^2 + (y^2)^2) = (x^2 - y^2)(x^4 + x^2y^2 + y^4)$

(b) $\dfrac{x^3}{8} - \dfrac{512}{x^3}$ (3)

Solution: $\dfrac{x^3}{8} - \dfrac{512}{x^3} = \left(\dfrac{x}{2}\right)^3 - \left(\dfrac{8}{x}\right)^3 = \left(\dfrac{x}{2} - \dfrac{8}{x}\right)\left(\left(\dfrac{x}{2}\right)^2 - \dfrac{x}{2}\cdot\dfrac{8}{x} - \left(\dfrac{8}{x}\right)^2\right) = \left(\dfrac{x}{2} - \dfrac{8}{2}\right)\left(\dfrac{x^2}{4} - 4 - \dfrac{64}{x^2}\right)$

3. Solve for x (4)

$$\frac{4}{x+2} + \frac{1}{x-2} = \frac{4}{x^2-4}$$

Solution:

$$\begin{aligned} \frac{4}{x+2} + \frac{1}{x-2} &= \frac{4}{x^2-4} \\ \frac{4(x-2)+(x+2)}{(x+2)(x-2)} &= \frac{4}{x^2-4} \\ \frac{4x-8+x+2}{(x+2)(x-2)} &= \frac{4}{x^2-4} \\ 5x-6 &= 4 \\ x &= 2 \end{aligned}$$

Checking, we cannot have $x = 2$ as a solution. Hence the equation has no solution

4. (a) Use the distance formula to show that the triangle with given vertices is an isosceles triangle; $A(-7,4)$, $B(-3,10)$ and $C(1,3)$. (4)

Solution: The length of AB is $\sqrt{(-7+3)^2+(4-10)^2} = \sqrt{16+36} = \sqrt{52}$.
The length of BC is $\sqrt{(-3-1)^2+(10-3)^2} = \sqrt{16+49} = \sqrt{65}$.
The length of AC is $\sqrt{(-7-1)^2+(4-3)^2} = \sqrt{64+1} = \sqrt{65}$.
Now since AB = AC then the triangle with the given vertices is isosceles

(b) Find the perimeter of the triangle that is formed by joining the midpoints of the three sides of triangle ABC in part a). (5)

Solution: Let D, E and F be the midpoints of AB, BC and AF respectively.
The coordinates of D are $\left(\frac{-7-3}{2}, \frac{4+10}{2}\right) = (-5,7)$.
The coordinates of E are $\left(\frac{-3+1}{2}, \frac{10+3}{2}\right) = \left(-1, \frac{13}{2}\right)$.
The coordinates of F are $\left(\frac{-7+1}{2}, \frac{4+3}{2}\right) = \left(-3, \frac{7}{2}\right)$.

Now the length of DE is $\sqrt{(-5-1)^2+(7-\frac{13}{2})^2} = \sqrt{36+\frac{1}{4}} = \frac{\sqrt{145}}{2}$.
The length of EF is $\sqrt{(-1+3)^2+(\frac{13}{2}-\frac{7}{2})^2} = \sqrt{4+9} = \sqrt{13}$.

The length of DF is $\sqrt{(-5+3)^2+(7-\frac{7}{2})^2} = \sqrt{4+\frac{49}{4}} = \frac{\sqrt{65}}{2}$.
Hence the perimeter is $\frac{\sqrt{145}}{2} + \sqrt{13} + \frac{\sqrt{65}}{2} = \frac{2\sqrt{13}+\sqrt{65}+\sqrt{145}}{2}$

5. (a) Indicate the following numbers $4\pi, 0, -\frac{2}{3}, 4.\overline{3}$ as natural, integers, rational and irrational. (4)

Solution:
4π irrational; $4.\overline{3}$ rational; 0 integer; $-\frac{2}{3}$ rational.

(b) Simplify the expression $\mid x-3 \mid + \mid x-4 \mid$ given that $3 < x < 4$. (3)

Solution: Since x is in the interval set $(3, 4)$
then the first expression $\mid x-3 \mid$ is positive
and $\mid x-4 \mid$ is negative
so we have $(x-3) + -(x-4)$
$= x - 3 - x + 4 = 1$

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6. (a) Determine whether $(\frac{3}{5}, \frac{-17}{5})$ lies on the graph of $y = \frac{-2}{3}x - 3$. (2)

Solution: Let $y = f(x)$. Then $f(\frac{3}{5}) = \frac{-2}{3} \times \frac{3}{5} - 3 = \frac{-17}{5}$. So the point $(\frac{3}{5}, \frac{-17}{5})$ lies on the graph.

(b) Determine whether the following lines are parallel, perpendicular, or neither.

i. $3x - 4y = 12$, $4x - 3y = 12$; (4)

Solution: Express the lines in the form $y = mx + c$. Thus first line is $y = \frac{3}{4}x - 3$ and the second one takes the form $y = \frac{4}{3}x - 4$. The slopes of the two lines are neither equal nor give -1 when multiplied. Thus the lines are neither parallel nor perpendicular.

ii. $3y + 48 = 15x$, $2y - 10x + 4 = 0$. (3)

Solution: The equations yield $y = 5x - 16$ and $y = 5x - 2$ respectively. Comparing the coefficients of x, the slopes are equal, hence the lines are parallel.

7. Simplify $x(x+y)^{-1} + y(x-y)^{-1}$ (4)

Solution: $x(x+y)^{-1} + y(x-y)^{-1}$
$= \frac{x}{x+y} + \frac{y}{x-y}$
$= \frac{x(x-y)+y(x+y)}{(x+y)(x-y)}$

$= \frac{x^2-xy+yx+y^2}{x^2-y^2}$
$= \frac{x^2+y^2}{x^2-y^2}$

8. Consider the equation

$$y = -\frac{1}{x^3} - x$$

Determine whether the graph of the equation is symmetric about

(a) the x axis (2)

Solution:
Replacing y with $-y$ yields $-y = -\frac{1}{x^3} - x$ which is not equivalent to the original equation. Hence the graph is not symmetric about the x axis

(b) the y axis (2)

Solution:
Replacing x with $-x$ yields $y = -\frac{1}{(-x)^3} - (-x) = \frac{1}{x^3} + x$ which is not equivalent to the original equation. Hence the graph is not symmetric about the y axis

(c) the origin (2)

Solution:
Replacing y with $-y$ and x with $-x$ yields $-y = \frac{1}{x^3} + x$ which (upon multiplying through by -1) is equivalent to the original equation. Hence the graph is symmetric about the origin

9. (a) The points $(7, -4)$ and $(-3, 4)$ are the endpoints of the diameter of a circle. Find the standard equation for the circle. (*Hint : The standard equation of a circle with center* (h, k) *and radius* r *is* $(x-h)^2 + (y-k)^2 = r^2$) (5)

Solution: The diameter is $d = \sqrt{(7-(-3))^2 + (-4-4)^2} = \sqrt{164} = 2\sqrt{41}$.
Therefore the radius of the circle is $r = \frac{d}{2} = \frac{2\sqrt{41}}{2} = \sqrt{41}$.
The centre of the of the circle is the midpoint of the diameter and so has coordinates $\left(\frac{7-3}{2}, \frac{-4+4}{2}\right) = (2, 0)$.
Hence the equation of the circle is

$$(x-2)^2 + (y-0)^2 = (\sqrt{41})^2$$

$$(x-2)^2 + y^2 = 41$$

(b) Determine the centre and radius for the circle with equation (5)

$$16x^2 + 16y^2 - 16x + 24y - 35 = 0$$

Solution: $16x^2 + 16y^2 - 16x + 24y - 35 = 0$

$\Rightarrow x^2 + y^2 - x + \frac{3}{2}y - \frac{35}{16} = 0$

$\Rightarrow x^2 - x + y^2 + \frac{3}{2}y - \frac{35}{16} = 0$

$\Rightarrow (x^2 - x + \left(-\frac{1}{2}\right)^2) + (y^2 + \frac{3}{2}y + \left(\frac{3}{4}\right)^2) = \left(-\frac{1}{2}\right)^2 + \left(\frac{3}{4}\right)^2 + \frac{35}{16}$

$\Rightarrow \left(x - \frac{1}{2}\right)^2 + \left(y + \frac{3}{4}\right)^2 = \frac{1}{4} + \frac{9}{16} + \frac{35}{16} = \frac{4+9+35}{16} = \frac{48}{16} = 3$

$\Rightarrow \left(x - \frac{1}{2}\right)^2 + \left(y - \left(-\frac{3}{4}\right)\right)^2 = (\sqrt{3})^2$

Hence the circle has centre $\left(\frac{1}{2}, -\frac{3}{4}\right)$ and radius $\sqrt{3}$.